JETHEX
EURO TALENT HUNT EXAM 2026

CLASS : X Moving to XI

euro
JEE – NEET

Time	**: 60 Minutes**
Marks	**: 240**
Medium	**: English**

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **ORIGINAL Copy** carefully with **blue/black ball point pen only**.

2. The test is of **90 minutes** duration and the Test Booklet contains **60 multiple-choice questions** (four options with a single correct answer) from **Physics, Chemistry, Biology, Mathematics, Mental Ability & Astronomy.**

3. Wherever the symbols/constants are not mentioned, they are to be considered as per their standard meaning/value.

4. Each question carries **4 marks**. For each correct response, the candidate will get **4 marks**.The **maximum marks are 240**.

5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.

6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.

7. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy)** to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.

8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.

9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.

10. Each candidate must show on-demand his/her Admit Card to the Invigilator.

11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.

12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an **Unfair Means case**.

13. Use of Electronic/Manual Calculator is prohibited.

14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination along with Public Examinations (Prevention of unfair means act 2024).

15. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.

16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : …….

Roll Number : in figures ……...

In words ……

Candidate's Signature : …… Invigilator's Signature : ……...

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PART I - PHYSICS

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

1. A car starts from rest and moves with uniform acceleration for 10 s, covering 150 m. It then continues with the velocity acquired for another 20 s. What is the total distance covered in 30 s?

(A) 450 m (B) 600 m
(C) 750 m (D) 900 m

2. A 5 kg block is pulled on a horizontal surface by a horizontal force of 30 N. If it accelerates at 2 m/s², what is the magnitude of the frictional force acting on it?

(A) 10 N (B) 20 N
(C) 30 N (D) 40 N

3. Two objects of masses m and 4m are separated by a distance r. If the distance is doubled and the mass 4m is replaced by 8m, the gravitational force between them becomes:

(A) Half of the original (B) Equal to the original
(C) Twice the original (D) Four times the original

4. A sound wave travels from air into water. Which statement is correct?

(A) Its frequency decreases and wavelength decreases
(B) Its frequency remains unchanged and wavelength increases
(C) Its frequency increases and speed remains unchanged
(D) Its frequency and wavelength both remain unchanged

5. A machine lifts a 200 kg load vertically through 5 m in 10 s. Taking g = 10 m/s², what is the minimum power delivered to the load?

(A) 500 W (B) 1000 W
(C) 2000 W (D) 10,000 W

6. An object is placed 30 cm in front of a concave mirror of focal length 20 cm. The image formed is:

(A) Virtual, erect and magnified
(B) Real, inverted and magnified
(C) Real, inverted and diminished
(D) Real, inverted and same size

7. A ray of light passes from glass (refractive index 1.5) to air at an angle of incidence greater than the critical angle. Which phenomenon occurs?

(A) Ordinary refraction only
(B) Dispersion
(C) Total internal reflection
(D) Scattering

8. A student can see nearby objects clearly but cannot see the blackboard clearly from the back of the classroom. The defect and suitable corrective lens are:

(A) Hypermetropia; convex lens
(B) Myopia; concave lens
(C) Presbyopia; convex lens
(D) Myopia; convex lens

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9. White light passes through a prism and forms a spectrum. Which colour is deviated the most, and why?

(A) Red, because it travels slowest in glass

(B) Violet, because glass has the highest refractive index for it

(C) Red, because it has the shortest wavelength

(D) Violet, because its frequency changes inside glass

10. A 2 kg body moving at 6 m/s is brought to rest by a constant opposing force over a distance of 3 m. The magnitude of the force is:

(A) 6 N (B) 12 N

(C) 18 N (D) 24 N

PART I - CHEMISTRY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

11. A compound contains 40% carbon, 6.67% hydrogen and 53.33% oxygen by mass. Its simplest whole-number atomic ratio C:H:O is:

(A) 1:1:1 (B) 1:2:1

(C) 2:4:2 (D) 2:1:2

12. An atom has electronic configuration 2, 8, 7. Which statement is most appropriate?

(A) It tends to lose one electron and form a +1 ion

(B) It tends to gain one electron and form a −1 ion

(C) It has valency 7 and is chemically inert

(D) It tends to lose seven electrons and form a +7 ion

13. Which equation represents both a displacement reaction and a redox reaction?

(A) $CaO + H_2O \rightarrow Ca(OH)_2$

(B) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

(C) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

(D) $CaCO_3 \rightarrow CaO + CO_2$

14. A solution has pH 3. Another solution has pH 5. Compared with the second solution, the first has approximately:

(A) 2 times higher H^+ concentration

(B) 10 times higher H^+ concentration

(C) 100 times higher H^+ concentration

(D) 1000 times higher H^+ concentration

15. Which salt is expected to give a basic solution in water?

(A) NaCl (B) NH_4Cl

(C) Na_2CO_3 (D) $CuSO_4$

16. Metal X reacts vigorously with cold water, while metal Y reacts with steam but not with cold water. Which order of reactivity is most likely?

(A) Y > X > H (B) X > Y > H

(C) H > X > Y (D) X > H > Y

17. An iron nail is placed separately in $CuSO_4$ and $ZnSO_4$ solutions. Which observation is correct?

(A) Iron displaces both Cu and Zn

(B) Iron displaces Cu but not Zn

(C) Iron displaces Zn but not Cu

(D) No displacement occurs in either solution

18. During electrolysis of molten sodium chloride, sodium metal is obtained at the cathode. This is because sodium ions:

(A) Gain electrons and are reduced
(B) Lose electrons and are oxidized
(C) Gain protons and are reduced
(D) Lose protons and are oxidized

19. Which pair of substances would produce a salt and water but no gas under ordinary acid–base neutralization conditions?

(A) $HCl + Na_2CO_3$ (B) $H_2SO_4 + Zn$
(C) $HNO_3 + KOH$ (D) $HCl + NaHCO_3$

20. A metal oxide reacts with both HCl and NaOH. The oxide is most likely:

(A) Na_2O (B) MgO
(C) Al_2O_3 (D) CO_2

PART I - BIOLOGY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

21. A farmer crosses two pure-breeding pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy). The F1 plants are selfed. Among 1600 F2 offspring, approximately how many are expected to have wrinkled yellow seeds?

(A) 100 (B) 300
(C) 900 (D) 600

22. In a food chain, grass → grasshopper → frog → snake, the grass fixes 80,000 kJ of energy. Assuming the usual 10% transfer rule, how much energy is expected to reach the snake?

(A) 8000 kJ (B) 800 kJ
(C) 80 kJ (D) 8 kJ

23. Two heterozygous tall pea plants are crossed (Tt × Tt). A randomly selected offspring is known to be tall. What is the probability that it is heterozygous?

(A) ¼ (B) 1/2
(C) 2/3 (D) 3/4

24. During vigorous exercise, a person breathes faster even though the oxygen concentration of incoming air is unchanged. What is the most direct physiological trigger?

(A) Increased carbon dioxide and H+ in blood
(B) Increased glucose in plasma
(C) Lowered nitrogen in inhaled air
(D) Reduced red blood cell count within seconds

25. A healthy plant is kept in waterlogged soil for several days. Its leaves wilt even though the soil is wet. Which explanation is strongest?

(A) Oxygen shortage in root cells reduces active mineral uptake
(B) Xylem begins carrying sugar instead of water
(C) Stomata permanently convert to lenticels
(D) Transpiration necessarily increases to zero resistance

26. A human pancreatic duct is blocked but blood supply to the gland remains normal. Which process would most directly be impaired?

(A) Release of insulin into blood
(B) Delivery of digestive enzymes into the small intestine
(C) Filtration of nitrogenous wastes

(D) Transport of oxygen by haemoglobin

27. A student observes that a plant shoot bends toward a one-sided light source. Which sequence best explains the curvature?

(A) Auxin accumulates on shaded side → greater cell elongation there
(B) Auxin accumulates on lit side → greater cell elongation there
(C) Auxin is destroyed everywhere → shoot contracts
(D) Gibberellin on the lit side stops all growth

28. The filtrate entering the proximal convoluted tubule of a healthy nephron contains glucose. In a healthy individual, why is glucose generally absent from final urine?

(A) Glucose never passes through glomerular capillaries
(B) It is selectively reabsorbed into blood from the tubule
(C) It is converted to urea in Bowman's capsule
(D) It leaves the kidney through the renal artery without filtration

29. An animal muscle cells are forced to produce ATP with inadequate oxygen during a sprint. Which change is most likely?

(A) Lactic acid builds up and ATP yield per glucose drops
(B) Carbon dioxide fixation increases
(C) Mitochondrial aerobic oxidation accelerates without oxygen
(D) Glucose stops breaking down altogether

30. A pesticide kills most insects in an agricultural field, including pollinators. Which outcome best illustrates an indirect ecological impact?

(A) All plants immediately stop respiration
(B) Seed formation in insect-pollinated crops may decline
(C) Herbivores instantly become producers
(D) Energy transfer becomes 100% efficient

PART I - MATHEMATICS

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

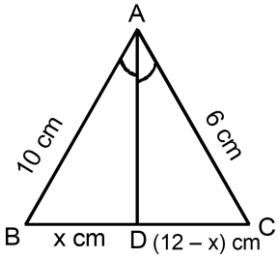
31. The roots of the quadratic equation $x^2 - 2\sqrt{3}x - 22 = 0$ are:

(A) imaginary
(B) real, rational, equal
(C) real, rational, unequal
(D) real, irrational, unequal

32. When $x^{40} + 2$ is divided by $x^4 + 1$, what is the remainder?

(A) 1 (B) 2
(C) 3 (D) 4

33. In a △ABC, AD is the bisector of ∠A, meeting side BC at D. If AB = 10 cm, AC = 6 cm, BC = 12 cm, find BD.



(A) 3.3 (B) 18

(C) 7.5 (D) 1.33

34. Euclid's division Lemma states that if a and b are any two positive integers, then there exist unique integers q and r such that -

(A) a = bq + r, 0 < r ≤ b

(B) a = bq + rm 0 ≤ q < b

(C) a = bq + r, 0 ≤ r < b

(D) a = bq + r, 0 < q ≤ b

35. If the arithmetic mean of 5, 7, 9, x is 9 then the value of x is:

(A) 11 (B) 15

(C) 18 (D) 16

36. If $\cos(40° + x) = \sin 30°$, then x is equal to

(A) 20° (B) 30°

(C) 60° (D) 0° △ Ap

37. The sum of all 2- digit numbers is –

(A) 4750 (B) 4905

(C) 3776 (D) 4680

38. The point whose abscissa is equal to its ordinate and which is equidistant from $A(-1,0)$ and $B(0,5)$ is

(A) $(1,1)$ (B) $(2,2)$

(C) $(-2,-2)$ (D) (3, 3)

39. If the medians of two equilateral triangles are in the ratio 3: 2, then what is the ratio of their sides?

(A) 1: 1 (B) 2: 3

(C) 3: 2 (D) $\sqrt{3}:\sqrt{2}$

40. If the distance f:om the vertex to the centroid of an equilateral triangle is 6 cm, then what is the area of the triangle?

(A) 24cm^2 (B) $27\sqrt{3}\text{cm}^2$

(C) 12cm^2 (D) $12\sqrt{3}\text{cm}^2$

PART II – MENTAL ABILITY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

41. Find the next term: 3, 8, 18, 38, 78, __.

(A) 118 (B) 148

(C) 158 (D) 168

42. In a code, each letter is advanced by its position in the word (1st by 1, 2nd by 2, and so on). What is the code for PLANE?

(A) QNDRJ (B) QNCRJ

(C) QNDSI (D) RMDSJ

43. A cube is painted on all six faces and then cut into 125 equal small cubes. How many small cubes have exactly two painted faces?

(A) 24 (B) 36

(C) 48 (D) 54

44. Five students P, Q, R, S, T stand in a row facing north. T is at the far left; S is at the far right; Q is immediately right of P; R is immediately left of P. Who is in the middle?

(A) P (B) Q

(C) R (D) S

45. A person walks 6 km east, 8 km north, 6 km west and 2 km south. What are the magnitude and direction of the displacement?

(A) 6 km north (B) 8 km south
(C) 2 km north (D) 14 km north

46. How many 3-digit even numbers can be formed without repetition from digits 1, 2, 3, 4, 5?

(A) 12 (B) 18
(C) 24 (D) 30

47. In a group of 80 students, 46 like Physics, 39 like Chemistry and 15 like neither. How many like both subjects?

(A) 5 (B) 10
(C) 20 (D) 25

48. If 1 January of a non-leap year is a Monday, what day of the week is 1 March?

(A) Wednesday (B) Thursday
(C) Friday (D) Saturday

49. Find the missing number in the sequence: 2, 6, 12, 20, 30, 42, __.

(A) 52 (B) 54
(C) 56 (D) 60

50. Statements: All astronomers are scientists. Some scientists are artists. Conclusions: I. Some astronomers are artists. II. All astronomers are scientists. Which follows?

(A) Only I (B) Only II
(C) Both I and II (D) Neither I nor II

PART II – ASTRONOMY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

51. A planet travels in an elliptical orbit. Its speed is greatest at perihelion mainly because:

(A) Gravitational force vanishes at aphelion
(B) Equal areas are swept in equal times, requiring faster motion nearer the Sun
(C) Its mass increases near the Sun
(D) Orbital radius remains fixed while speed changes

52. A planet orbits a star at a mean distance four times Earth's mean Sun-distance. If the star has the Sun's mass, its orbital period is approximately:

(A) 2 years (B) 4 years
(C) 8 years (D) 16 years

53. Two stars have identical surface temperature, but star X has 9 times the radius of star Y. Assuming both radiate approximately as blackbodies, which is correct?

(A) X has 3 times the luminosity
(B) X has 9 times the luminosity
(C) X has 81 times the luminosity
(D) Luminosities are equal

54. A star spectrum shows absorption lines shifted toward longer wavelengths relative to laboratory values. The most direct inference is:

(A) The star is approaching Earth
(B) The star is moving away from Earth
(C) The star contains no hydrogen
(D) The star is not emitting light

55. A telescope objective has a diameter twice that of another telescope. Ignoring losses, how does its light-gathering power compare?

(A) Twice
(B) Four times
(C) Eight times
(D) Sixteen times

56. Which observation best supports the conclusion that the Moon has synchronous rotation?

(A) The Moon has no gravity
(B) Roughly the same lunar hemisphere faces Earth over time
(C) The lunar orbit is perfectly circular
(D) The Moon never revolves around Earth

57. At local solar noon on an equinox, an observer at latitude 20° N measures the Sun's altitude. Ignoring refraction, approximately what is it?

(A) 20°
(B) 70°
(C) 90°
(D) 110°

58. A galaxy spectral lines exhibit a larger redshift than those of a nearby galaxy. Under the usual cosmological interpretation, this suggests it:

(A) Is necessarily smaller
(B) Is generally receding faster
(C) Has no stars
(D) Must be within the Solar System

59. An electromagnetic wave of frequency 6 × 10^14 Hz travels through vacuum. Using c = 3 × 10^8 m/s, its wavelength is:

(A) 5 × 10^-7 m
(B) 2 × 10^-6 m
(C) 5 × 10^-5 m
(D) 2 × 10^-7 m

60. Which statement correctly describes an ISRO launch vehicle placing a satellite into Earth orbit?

(A) A satellite needs continuous engine thrust to remain in stable orbit
(B) The satellite stays up because there is no gravity in orbit
(C) A suitable horizontal speed allows gravitational acceleration to provide centripetal acceleration
(D) Once launched, the satellite travels in a straight line without acceleration

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ANSWER KEY

CLASS X | ETHEX 2027 SAMPLE TEST PAPER

PHYSICS

1. C	**2. B**	**3. A**	**4. B**	**5. B**	**6. B**	**7. C**	**8. B**	**9. B**	**10. B**

CHEMISTRY

11. B	**12. B**	**13. C**	**14. C**	**15. C**	**16. B**	**17. B**	**18. A**	**19. C**	**20. C**

BIOLOGY

21. B	**22. C**	**23. C**	**24. A**	**25. A**	**26. B**	**27. A**	**28. B**	**29. A**	**30. B**

MATHEMATICS

31. D	**32. C**	**33. C**	**34. C**	**35. B**	**36. A**	**37. B**	**38. B**	**39. C**	**40. B**

IQ / MENTAL ABILITY

41. C	**42. A**	**43. B**	**44. A**	**45. A**	**46. C**	**47. C**	**48. B**	**49. C**	**50. B**

ASTRONOMY

51. B	**52. C**	**53. C**	**54. B**	**55. B**	**56. B**	**57. B**	**58. B**	**59. A**	**60. C**

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